

10.7.82

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Question

Find the coordinates of the point at which the circles $x^2 + y^2 - 4x - 2y = -4$ and $x^2 + y^2 - 12x - 8y = -36$ touch each other. Also find equations of the common tangents touching the circles in the distinct points.

Theoretical Solution

Circle (C_1):

$$x^2 + y^2 - 4x - 2y + 4 = 0 \quad (1)$$

$$\implies u = \begin{pmatrix} -2 \\ -1 \end{pmatrix} \text{ and } f = 4 \quad (2)$$

$$\implies \text{Centre}(\mathbf{c}_1) = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \text{ and } r_1 = \sqrt{\|u\|^2 - f} = 1 \quad (3)$$

Circle (C_2):

$$x^2 + y^2 - 12x - 8y + 36 = 0 \quad (4)$$

$$\implies u = \begin{pmatrix} -6 \\ -4 \end{pmatrix} \text{ and } f = 36 \quad (5)$$

$$\implies \text{Centre}(\mathbf{c}_2) = \begin{pmatrix} 6 \\ 4 \end{pmatrix} \text{ and } r_2 = \sqrt{\|u\|^2 - f} = 4 \quad (6)$$

Theoretical Solution

This point of contact(**P**) lies on the line segment connecting the two centers, **c**₁ and **c**₂, and divides it in the ratio of their radii, $r_1 : r_2$.

$$\mathbf{P} = \frac{1}{r_1 + r_2} \begin{pmatrix} c_1 & c_2 \end{pmatrix} \begin{pmatrix} r_2 \\ r_1 \end{pmatrix} \quad (7)$$

$$\mathbf{P} = \frac{1}{5} \begin{pmatrix} 2 & 6 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 4 \\ 1 \end{pmatrix} \quad (8)$$

$$\mathbf{P} = \begin{pmatrix} 14/5 \\ 8/5 \end{pmatrix} \quad (9)$$

Theoretical Solution

Let $\mathbf{n}^\top \mathbf{x} = c$ be the equation of common tangent to both the circles.

$$\frac{|\mathbf{n}^\top \mathbf{c}_1 - c|}{\|\mathbf{n}\|} = r_1 \implies \mathbf{n}^\top \mathbf{c}_1 = c \pm \|\mathbf{n}\| \quad (10)$$

$$\frac{|\mathbf{n}^\top \mathbf{c}_2 - c|}{\|\mathbf{n}\|} = r_2 \implies \mathbf{n}^\top \mathbf{c}_2 = c \pm 4 \|\mathbf{n}\| \quad (11)$$

$$\mathbf{n}^\top (\mathbf{c}_2 - \mathbf{c}_1) = \pm 3 \|\mathbf{n}\| \quad (12)$$

Theoretical Solution

substitute $\mathbf{n} = \begin{pmatrix} -m \\ 1 \end{pmatrix}$ where m is the slope of the line.

$$\Rightarrow \begin{pmatrix} -m & 1 \end{pmatrix} \begin{pmatrix} 4 \\ 3 \end{pmatrix} = \pm 3 \sqrt{1 + m^2} \quad (13)$$

$$\Rightarrow 16m^2 - 24m + 9 = 9(1 + m^2) \quad (14)$$

$$\Rightarrow 7m^2 - 24m = 0 \quad (15)$$

$$m = 0 \text{ and } m = 24/7 \Rightarrow \mathbf{n} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ and } \mathbf{n} = \begin{pmatrix} -24 \\ 7 \end{pmatrix} \quad (16)$$

Theoretical Solution

To find the tangent line, substitute $\mathbf{n} = \begin{pmatrix} -24 \\ 7 \end{pmatrix}$ in equation (10), we get

$$c = -66 \text{ and } c = -16 \quad (17)$$

And equation (11) holds only when $c = -16$.

Therefore the two tangent lines are $\begin{pmatrix} 0 & 1 \end{pmatrix} \mathbf{x} = 0$ and $\begin{pmatrix} -24 & 7 \end{pmatrix} \mathbf{x} = -16$

